Unit: 6 Unit 2: Exponents Radicals **Name:** ________________
Topic: Rationalizing Denominator **Date:** ___________

Homework (Carolina Reaper)

Q1. A painter mixes $2\cdot\sqrt{3}$ liters of blue paint with $5\cdot\sqrt{2}$ liters of white paint. What is the ratio of blue paint to total paint, with a rationalized denominator?

(A) $\frac{2\cdot\sqrt{3}}{5\cdot\sqrt{2}+2\cdot\sqrt{3}}$

(B) $\frac{2\cdot\sqrt{3}\cdot\sqrt{2}}{5\cdot2+2\cdot\sqrt{3}\cdot\sqrt{2}}$

(C) $\frac{4\cdot\sqrt{6}}{10+4\cdot\sqrt{6}}$

(D) $\frac{4\cdot\sqrt{6}}{10+4\cdot\sqrt{6}}\cdot\frac{10-4\cdot\sqrt{6}}{10-4\cdot\sqrt{6}}$

(E) $\frac{2\cdot\sqrt{3}\cdot\sqrt{2}}{5\cdot\sqrt{2}+2\cdot\sqrt{3}\cdot\sqrt{2}}$

Q2. A musician plays a rhythm with a frequency of $\frac{1}{\sqrt{5}}$ Hz. What is this frequency with a rationalized denominator?

(A) $\frac{1}{\sqrt{5}}$

(B) $\frac{\sqrt{5}}{5}$

(C) $\frac{1}{5}$

(D) $\frac{\sqrt{5}}{25}$

(E) $\frac{1}{25}$

Q3. The area of a rectangular canvas is $4\cdot\sqrt{3}$ square meters. If its length is $2\cdot\sqrt{2}$ meters, what is its width with a rationalized denominator?

(A) $\frac{4\cdot\sqrt{3}}{2\cdot\sqrt{2}}$

(B) $\frac{2\cdot\sqrt{3}\cdot\sqrt{2}}{2}$

(C) $\frac{4\cdot\sqrt{6}}{4}$

(D) $\frac{4\cdot\sqrt{3}}{2\cdot\sqrt{2}}\cdot\frac{\sqrt{2}}{\sqrt{2}}$

(E) $\sqrt{6}$

Q4. A sound wave has a frequency of $\frac{1}{2\cdot\sqrt{7}}$ Hz. What is this frequency with a rationalized denominator?

(A) $\frac{1}{2\cdot\sqrt{7}}$

(B) $\frac{\sqrt{7}}{14}$

(C) $\frac{1}{14}$

(D) $\frac{\sqrt{7}}{28}$

(E) $\frac{1}{28}$

Q5. A painter needs $\frac{3}{\sqrt{11}}$ liters of paint to cover a wall. What is the amount of paint needed with a rationalized denominator?

(A) $\frac{3}{\sqrt{11}}$

(B) $\frac{3\cdot\sqrt{11}}{11}$

(C) $\frac{3\cdot\sqrt{11}}{121}$

(D) $\frac{3}{11}$

(E) $\frac{3\cdot\sqrt{11}}{11}$

Q6. The length of a string is $5\cdot\sqrt{2}$ meters. If it is divided into $\frac{1}{\sqrt{3}}$ parts, what is the length of each part with a rationalized denominator?

(A) $\frac{5\cdot\sqrt{2}}{\sqrt{3}}$

(B) $\frac{5\cdot\sqrt{2}\cdot\sqrt{3}}{3}$

(C) $\frac{5\cdot\sqrt{6}}{3}$

(D) $\frac{5\cdot\sqrt{2}}{\sqrt{3}}\cdot\frac{\sqrt{3}}{\sqrt{3}}$

(E) $\frac{5\cdot\sqrt{6}}{3}$

Q7. A musician plays a song with a rhythm pattern of $\frac{2}{\sqrt{5}}$ beats per minute. What is this rhythm pattern with a rationalized denominator?

(A) $\frac{2}{\sqrt{5}}$

(B) $\frac{2\cdot\sqrt{5}}{5}$

(C) $\frac{2}{5}$

(D) $\frac{2\cdot\sqrt{5}}{25}$

(E) $\frac{2}{25}$

Q8. The area of a triangular canvas is $6\cdot\sqrt{3}$ square meters. If its base is $2\cdot\sqrt{2}$ meters, what is its height with a rationalized denominator?

(A) $\frac{6\cdot\sqrt{3}}{2\cdot\sqrt{2}}$

(B) $\frac{3\cdot\sqrt{3}\cdot\sqrt{2}}{2}$

(C) $\frac{3\cdot\sqrt{6}}{2}$

(D) $\frac{6\cdot\sqrt{3}}{2\cdot\sqrt{2}}\cdot\frac{\sqrt{2}}{\sqrt{2}}$

(E) $\frac{3\cdot\sqrt{6}}{2}$

Open Ended Questions (FRQ)

Q9. The length and width of a rectangular canvas are $3\cdot\sqrt{5}$ meters and $2\cdot\sqrt{7}$ meters respectively. Find the area of the canvas with a rationalized denominator.

Q10. A musician plays a song with a rhythm pattern of $\frac{3}{\sqrt{11}}$ beats per minute. What is this rhythm pattern with a rationalized denominator? Show all steps.

Chef's Tips

- Tip 1: Emphasize the importance of rationalizing denominators in real-world applications.
- Tip 2: Encourage students to use the conjugate to rationalize denominators.
- Tip 3: Remind students to simplify their answers after rationalizing denominators.